

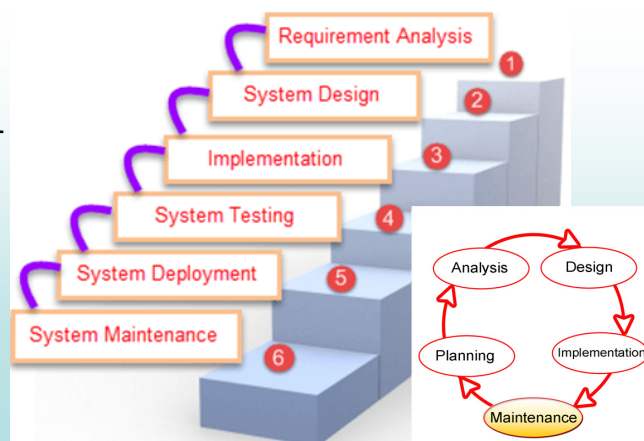
(Agile) Software Development

--- A Brief Introduction with Scrum

A Software Development Process

■ Waterfall Model:

- Distinct milestones with a development phase, a closing testing cycle;
- Rework of “perfect” plan creates corrective loops for adjustments;
- Communication and interfaces between different organizational units are a challenge in large distributed teams with a centralized project management.



Advantages and Disadvantages

- Advantages of the waterfall approach:
 - Most efficient when starting with *complete* requirements and coding with *error-free* design;
 - Thorough execution of the waterfall model produces a good set of documentation.
- Disadvantages:
 - “Perfect” plan expects *no changes*, which is impossible when carrying out larger/new projects;
 - ✓ Clients know better and more after trying out prototypes;
 - ✓ New requirements lead to design and coding changes;
 - ✓ Cost increases due to changes and contract constraints.

The (OO) Unified Process

- Key phases of software projects:
 - ***inception*** – understand the product under development
 - ***requirements*** - capture requirements (with clients)
 - ***design*** - create designs for (selected) requirements
 - ***implementation*** - develop software version for release
 - ***testing*** - test functionality and check for acceptability
- Inception is the initial short step of a project.
 - Methods:
 - Understand the product via requirement exploration
 - Objectives:
 - Define a good set, but not all the requirements
 - Generate a rough, but unreliable estimate
 - Create an initial plan, but not for the whole project

Software Requirements

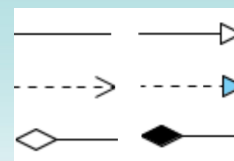
- Software requirements can be functional or non-functional.
 - Functional requirements describe what a system must do in terms of functionality.
 - They are described in use case documents.
 - A brief format has a title and a short description
 - An example:

Title	Navigation with arrow keys
Brief description	The user needs to navigate the virtual environment by using the four arrow keys.

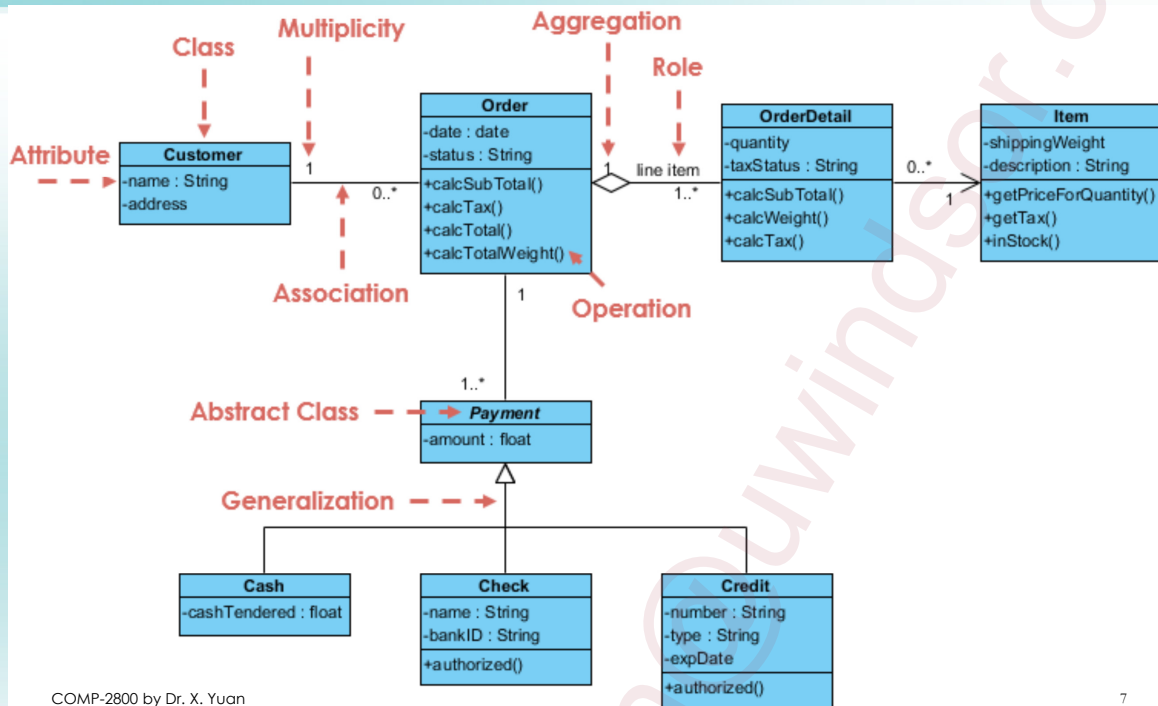
- Non-functional requirements specify how well the system performs the functionality
 - They are documents linked to specific use cases.
 - An example: response time in terms of milliseconds

Software Design

- It specifies the structure of a software by:
 - Deriving classes needed to fulfill requirements
 - Determining fields and methods for each class
 - Finding how the classes interact with each other
- UML class diagram
 - It is a static structure diagram that describes the system by showing its classes, including their names, attributes, methods, and relationships.
 - Symbols for visibility:
 - + (public), # (protected), - (private), / (derived)
 - Symbols for relationships:
 - Association, inheritance, dependency, realization, aggregation, composition



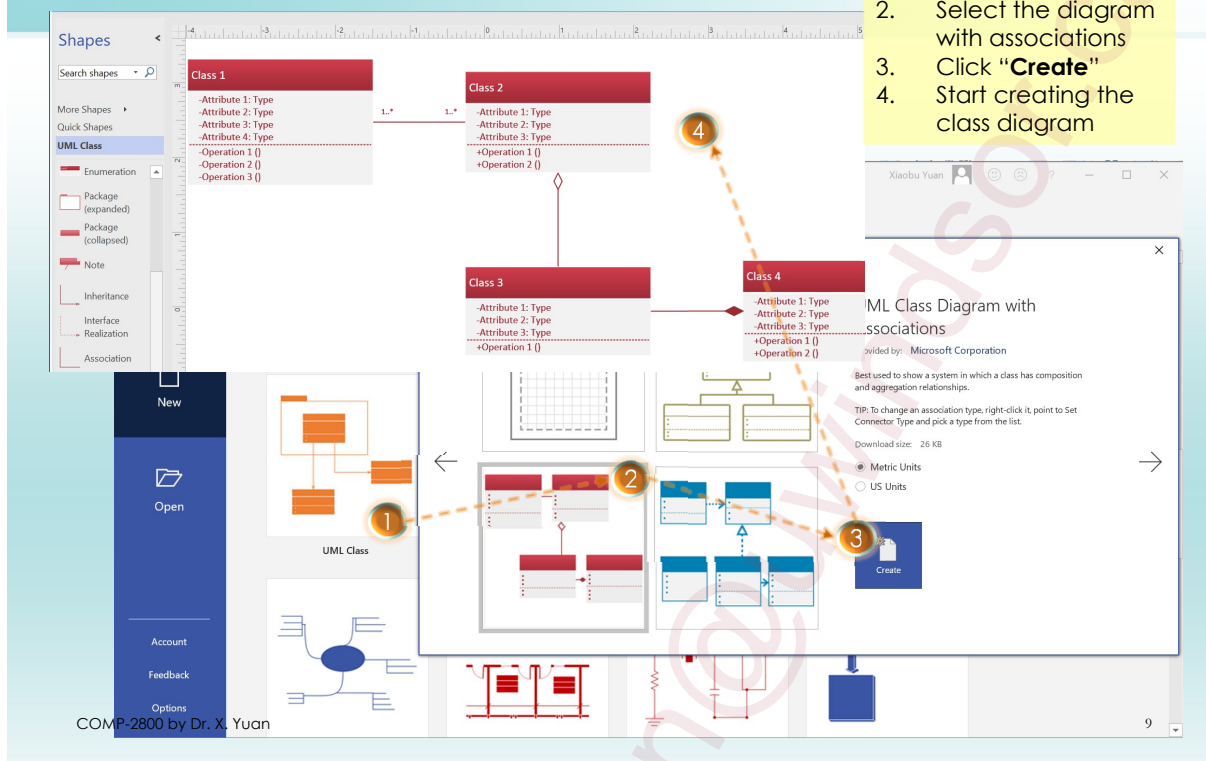
Class Diagram: Associations



Download Visio Pro 2019

1. Go to "www.cs.uwindsor.ca"
2. Select "Technical Resources ..."
3. Click "Software Download"
4. Search for "visio"
5. Click "Visio Professional 2019"
6. Install the software

Class Diagram with Visio



Software Implementation

- UP: an iterative process
 - Coding produces a working system
 - The availability of a working system helps to feed back ideas and observations.
 - Design on paper is not interactive and very difficult, if not impossible, to complete for the whole system.
 - Functionality accumulates over iterations until complete.
- The writing of actual codes
 - Coding starts with artifacts from software design
 - Extra work is usually required for more details.
 - In-depth understanding likely leads to:
 - Addition or change of classes and associations
 - Addition or change of type representations
 - Addition or change of methods and attributes

Software Test

- Testing is an activity to execute a system or component under specified conditions.
 - A *test case* is a set of inputs, execution conditions, and expected results for a particular objective.
 - ✓ A *test suite* is a collection of test cases, typically related by applying a particular testing technique.
 - ✓ A *test procedure* is the detailed instructions for the set-up, execution, and evaluation of results for a given test case.
- A successful testing is a test that helps to identify errors.

Scrum: Three Roles

- Scrum is an agile framework for developing, delivering, and sustaining complex products.
 - Agile methods encourage flexible responses to change with adaptive planning, evolutionary development, early delivery, and continual improvement.
- Three roles:
 - The *product owner* represents the interests of all the stakeholders, e.g. customers, marketing, etc.
 - The development team is responsible for developing and testing the project deliverable.
 - A *scrum master* is responsible for the scrum process.

Scrum: The Three Documents

- Product backlog
 - It gathers and prioritizes all the requirements in a list.
- Sprint backlog
 - A sprint is a small development iteration (~2 weeks).
 - ✓ It contains design, development, testing, documentation.
 - ✓ It produces a tested and stable deliverable to be handed over to the customer and put into production by him.
 - A sprint backlog includes the use cases for a sprint.
 - ✓ Requirements are prioritized by the product owner.
- Sprint results are the document about the use cases completed during a sprint.

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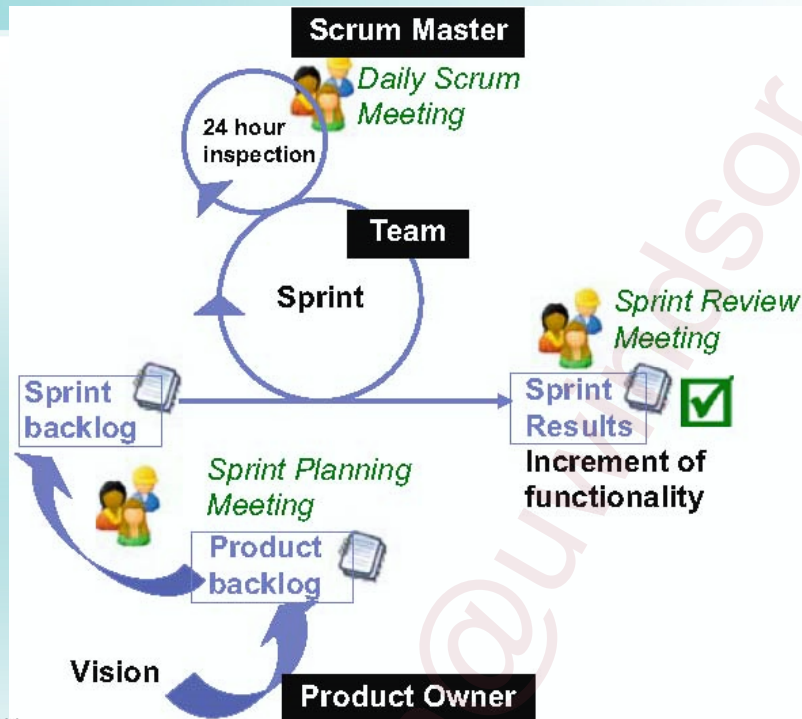
Scrum: The Three Meetings

- *Sprint planning meeting* before a sprint begins:
 - Decide what the team is going to deliver in a sprint.
 - a. Product owner presents the most important requirements;
 - b. The team selects the items to be delivered in the sprint.
- *Daily scrum meeting* for 15 minutes:
 - Determine progress and identify problems or issues.
 - ✓ Use *burndown charts* to show progress towards the goals.
 - ✓ Adjust the sprint backlog according to progress.
- *Sprint review meeting* at the end of the sprint:
 - The three parties meet to review, with some demos, the results achieved during the iteration.

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Scrum: The Relationships



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Scrum with Azure DevOps

1. Provide web address
2. Click "**Start free**" (free version for 5 or fewer)
3. Type in project name
4. Click "**Continue**"

The screenshot shows the Azure DevOps 'Get started with Azure DevOps' page. The steps are numbered as follows:

1. The URL bar shows the address: <https://azure.microsoft.com/en-us/services/devops/>.
2. The 'Start free' button is highlighted.
3. The 'Project name' field is filled with 'ScrumW21XY'.
4. The 'Continue' button is highlighted.

Additional details on the page include the Microsoft Azure logo, navigation links (Overview, Solutions, Products, Documentation, Pricing, Training), and a 'Contact Sales' link. The page title is 'Azure DevOps' with the subtitle 'Plan smarter, collaborate better, and ship faster with a modern dev services.' The 'Project visibility' is set to 'Private'. A checkbox for 'I would like to hear from Microsoft...' is present, and a 'Chat with Sales' button is at the bottom right.

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Prepare for One Sprint

1. Click "Invite" to add teammates
2. Click "Board"
3. Set Sprint's duration
4. Click "Capacity"
5. Select activity
6. Provide hours/day
7. Repeat Steps 5-6
8. Save changes when done

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Create A (Sprint) Backlog Item

You do not have any work scheduled yet
Schedule work from your product backlog or create new work items.

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Work in A Sprint

1. Stay in "Taskboard"
2. Click "+" to add task
3. Type in task title
4. Assign to teammate
5. Estimate effort (hour)
6. Change to "Doing"
7. Update remaining effort EVERY DAY!
8. Change to "Done" when task finished (watch the effort updating for the use case)

